

James Ridey

Software Engineer | Architecture, Performance & Reverse Engineering

Sydney, Australia • Australian citizen • Open to San Francisco Bay Area relocation

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Professional Experience

Nullify

Mar 2026 - present

Product Engineer

- **Architected and built the React/TypeScript frontend from the ground up**, with reusable components and CI-enforced dependency boundaries between primitives, features, and pages.
- Built generated API clients, typed error and retry handling, and Storybook visual-regression previews; added blocking browser smoke checks to catch failures before merge.
- Traced eager dependencies through source maps and module graphs; reduced gzipped startup assets from **932 kB to 453 kB** and added a CI size budget to prevent regressions.
- Built vulnerability-graph layouts using ELK in a background worker and **A* edge routing** to avoid node collisions; validated against 50 real findings.
- Cut Go Lambda initialisation from **1.9 s to 1.1 s** by removing unnecessary OpenAPI reflection (development deployment). Reused a Python subprocess to cut startup overhead from **9.7 s to 50 ms** on warm invocations.

Canva

2020 - 2025

Software Engineer

- Designed the state-management layer for Magic Design and delivered it two weeks ahead of its scheduled launch.
- Redesigning Marketplace filtering to reduce API response time and strengthened unit and integration tests during the refactor.

Google

2017 - 2019; two summer internships

Software Engineer Intern

- Ported Neighbourly, Google's local Q&A app, to KaiOS/JioPhone devices with **512 MB RAM**, resolving platform compatibility constraints; wrote the team's first porting specification.

ThoughtDesign Junior Software Engineer | 2014 - 2017

Selected Independent Engineering Projects

CUDA Seed Search Optimisation GPU profiling & parallel computation

- Profiled the search for the world seed behind Minecraft's pack.png image using NVIDIA Visual Profiler. Diagnosed early thread exits and split seed filtering and tree-position work into separate CUDA kernels; **roughly 2x performance** in local testing. Merged upstream.

JRB8 CPU & Toolchain Digital logic & Verilog

- Designed an 8-bit CPU, instruction set, and assembler; tested in simulation and on FPGA before Tiny Tapeout fabrication. Built a language, compiler, and tools to flash and run programs on the **working chip**.

Brother Embroidery Cartridge Hardware & software reverse engineering

- Traced PCB connections and the memory interface; built an Arduino reader and decoded stitch data using hex analysis and desktop-software reverse engineering.
- Built a working replacement cartridge; others reused the published code and findings for custom embroidery.

OnnxStream (C++): Five merged contributions, including SDXL Turbo support and tokenization/VAE scaling fixes.

Technical Skills & Credentials

Professional: Go, Python, TypeScript, JavaScript, AWS, React.

Independent projects: C/C++, CUDA, Verilog, STM32, Arduino, PCB design, reverse engineering.

Flare-On: Completed all challenges in 2022-2025; **157th of 4,100+** entrants in 2025.

Security: Burp Suite Certified Practitioner (2026); DEF CON Offensive IoT Exploitation training.

Education

Macquarie University | 2016 - 2020

BEng (Honours), Software Design. Head of Computing Award in five semesters.